Financial Markets: Intrinsic Dimensionality and Random Matrix Structure

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April 2026

1 Problem Statement

Present wisdom dictates that investors should diversify their portfolio by holding a variety of financial assets. However, underlying correlations between stocks and bonds may lead to insufficient diversification, even when many different assets are held. This phenomenon has direct consequences in risk management and portfolio construction and often leads to an underestimation of risk. Attempts to mitigate this issue examine the covariance matrix generated from asset returns. When applying this methodology to large portfolios or markets, the large number of assets creates a high-dimensional space. Random Matrix Theory (RMT) [1] addresses this high-dimensionality and is often used to parse the noise in financial returns, improving empirical estimates of covariance matrices. The result is that a subset of factors are the primary drivers of broader market behavior and that many covariates can be explained as spurious correlations.

This project studies the intrinsic dimensionality of S&P 500 returns. The goal is to estimate this lower-dimensional market structure and determine how it changes across time and market regimes. Specifically, the project estimates market dimensionality using three methods: Random Matrix Theory, the participation ratio, and the Levina-Bickel nearest neighbor estimator. The project also examines whether Random Matrix Theory can produce a cleaner covariance matrix for portfolio construction and economic factor interpretation. This follows the proposal's goal of identifying meaningful market factors, measuring dimensionality through time, and evaluating robustness across regimes.

2 Methodology

The analysis uses CRSP-derived S&P 500 return data, constituent membership history, and ticker-to-company mappings all from WRDS. For each analysis window, the code selects the actual S&P 500 constituents as of the window end date, which helps avoid survivorship bias. Assets with missing returns inside the window are dropped, and the return matrix is de-meaned before computing the covariance matrix.

Let $X \in \mathbb{R}^{n \times p}$ denote the demeaned return matrix, where n is the number of trading days and p is the number of assets. The empirical covariance matrix is

$$\Sigma = \frac{1}{n-1} X^\top X.$$

The covariance matrix is decomposed as

$$\Sigma = Q \Lambda Q^\top,$$

where Λ contains the eigenvalues and Q contains the eigenvectors.

2.1 Random Matrix Theory

Random Matrix Theory is the study of matrices with random variable entries. It establishes the statistical properties of eigenvalues and eigenvectors as the matrix dimensions become arbitrarily large. The field provides a baseline for how a matrix composed of entries that result purely from noise should behave under eigenvalue decomposition. In finance, asset returns are often treated as a random variable. As such, comparing the eigenvalues of the covariance matrix with the expected eigenvalues as dictated by RMT is a useful method for arguing the statistical significance of asset return behaviors.

The Marchenko-Pastur (MP) law offers a convenient approach for devising confidence intervals over eigenvalues. In the instance that eigenvalues are greater than the upper bound, then they are both practically and statistically significant. Likewise, those below the lower bound are statistically significant but practically insignificant. In the context of this project, the following procedure was applied:

1. The first dimensionality estimate uses the Marchenko–Pastur upper edge. Let,

$$q = \frac{p}{n}.$$

2. After estimating σ^2 from the noise bulk, as described in section 2.6, the Marchenko–Pastur bounds are:

$$\lambda_\pm = \sigma^2 (1 \pm \sqrt{q})^2.$$

3. Eigenvalues above λ_+ are counted as statistically significant market factors. Thus,

$$D_{\text{RMT}} = \#\{\lambda_i : \lambda_i > \lambda_+\}.$$

2.2 Participation Ratio

The participation ratio gives a continuous estimate of effective dimensionality:

$$D_{\mathrm{PR}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}.$$

This measures how evenly variance is distributed across eigenmodes.

2.3 Levina–Bickel Estimator

The traditional approaches to dimensionality reduction include principal component analysis (PCA) and multidimensional scaling (MDS). The former relies upon linear projection of the data, which destroys nonlinear relationships, and the latter, while preserving nonlinear relationships, is most often used for visualizing data. The need for further development of nonlinear dimensionality reduction techniques is the basis of the 2004 paper by Elizaveta Levina and Peter Bickel [2].

Similar to the methodology in the ISOMAP algorithm, the aforementioned paper leverages local linear embeddings to traverse the manifold and establish a measure of distance that preserves globally nonlinear relationships. The basic premise is as follows:

Given observed data points $X_1, \ldots, X_n$ that exist in a high-dimensional space $\mathbb{R}^p$. It is assumed that the m-dimensional manifold embedded via $g : \mathbb{R}^m \to \mathbb{R}^p$, a smooth mapping. The latent variables $Y_i \in \mathbb{R}^m$ are drawn i.i.d. from a smooth density f, and the observed data satisfied $X_i = g(Y_i)$. Because g is locally smooth and preserves its neighborhood, the local geometry of X_i reflects the intrinsic dimensionality of m. To determine m, a point, x, is fixed, and a sphere, $S_x(R)$, of radius R is centered on said point. Within the sphere, the density $f(x)$ is approximately constant.

From these priors, the mapped points behave like a homogeneous Poisson process on an m-dimensional ball. The authors define a counting process as:

$$N(t, x) = \sum_{i=1}^{n} \mathbf{1}\{X_i \in S_x(t)\}$$

Which records the number of neighbors within distance t. From this counting process, the arrival rate of data on the surface area of the sphere with respect to t is:

$$\lambda(t) = f(x)\, V(m)\, m\, t^{m-1}$$

Where the surface area of the sphere $S_x(t)$ is:

$$\frac{d}{dt}[\, V(m)\, t^m \,] = V(m)\, m\, t^{m-1}$$

Finally, setting $\theta = log(f(x))$, the log-likelihood of the observed process $N(t)$ is:

$$L(m, \theta) = \int_0^R \log \lambda(t)\, dN(t) \; - \; \int_0^R \lambda(t)\, dt$$

The authors continue to take the MLE of this log-likelihood and derive the estimator used in this project. Said estimator identifies the nonlinear, intrinsic dimension from nearest-neighbor distances. For each observation x_i, let $T_j(x_i)$ be the distance to its jth nearest neighbor. Then,

$$\hat{m}_k(x_i) = \left[\frac{1}{k-1} \sum_{j=1}^{k-1} \log \left(\frac{T_k(x_i)}{T_j(x_i)} \right) \right]^{-1}.$$

The final estimate averages this quantity across all valid observations.

2.4 Rolling-Window Analysis

The three estimators are computed across rolling windows to study how market dimensionality evolves over time. This allows the project to compare normal market periods with stressed regimes such as the 2008 financial crisis and the COVID-19 market shock.Moreover, rolling window analysis provides the opportunity to compare the behaviors of each estimator within a given regime. This opens the door to future analysis regarding joint behaviors of estimators and the percent changes of estimated values throughout time.

One of the objectives of this analysis is to see if the model can identify unknown market conditions. For example, if estimators behave a certain way during known recessions, then in the instance of an unknown recession, they may be useful for justifying portfolio decisions that assume a stressed market regime.

2.5 RMT-Cleaned Covariance and Portfolio Construction

The project also constructs an RMT-cleaned covariance matrix by replacing noise eigenvalues with their average while preserving the signal eigenvalues:

$$\tilde{\lambda}_i = \begin{cases} \lambda_i, & \lambda_i > \lambda_+, \\ \bar{\lambda}_{\text{noise}}, & \lambda_i \leq \lambda_+. \end{cases}$$

The cleaned covariance matrix is reconstructed as

$$\hat{\Sigma}_{\text{clean}} = Q \tilde{\Lambda} Q^\top.$$

Then, this cleaned covariance matrix is compared to the raw covariance matrix through global minimum-variance portfolios:

$$w^\star = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}.$$

2.6 Noise Variance Estimation

The Marchenko–Pastur upper edge depends on the noise variance σ^2, which must be estimated from the empirical spectrum. We compare three estimators: (i) the median of the bottom half of the spectrum (a common rule of thumb but biased low when $q \approx 1$); (ii) an iterative trimming procedure that initializes σ^2 as the mean of all eigenvalues, drops those exceeding the implied λ_+, and recomputes the mean of the remainder until convergence; and (iii) a least-squares fit of σ^2 to the empirical bulk histogram against the theoretical MP density. The iterative procedure is taken as canonical for all downstream analysis. To guard against rank deficiency in $q > 1$ windows and bootstrap samples, the estimator is restricted to strictly positive eigenvalues.

2.7 Robustness

Robust estimators of intrinsic dimensionality are consistent under mild perturbations of market data. In order to assess the quality of the three methods described in sections 2.1, 2.2, and 2.3, two stress tests were applied to the single-window estimates. The first method is asset sub-sampling, which draws 80% of assets without replacement, recomputes all three estimators, and repeats $B = 100$ times to measure sensitivity to the asset universe.

The second, more intensive approach is to implement moving-block bootstrap to construct confidence intervals around the estimates. Given the presence of distribution shifts in time series data, it is important to preserve short-range autocorrelation and volatility clustering. To accomplish this goal, data is divided into smaller contiguous elements, also referred to as blocks, and resampled with replacement. Furthermore, in an effort to maintain stationarity in the resampled data, this project employed a moving-block bootstrap, which allows time blocks to overlap. Specifically, the data was separated into contiguous 20-day moving blocks, resampled along the time axis, and repeated $B = 200$ times. The 95% percentile intervals were reported and produced graphically 2.

2.8 Eigenportfolios

Each eigenvector q_i is L1-normalized to produce a portfolio with unit gross exposure, $w_i = q_i / \sum_j |q_{ij}|$. The realized return of the i-th eigenportfolio is $r_i(t) = X(t)w_i$. By PCA orthogonality, q_1 is approximately a same-sign vector (a long-only "market mode"), while $q_2, \ldots, q_5$ have weights that sum to zero by construction, producing market-neutral long-short portfolios.

3 Results

3.1 Single-Window Estimates (2018–2019)

A two-year baseline ending 2019-12-31 contains $n = 503$ trading days and $p = 489$ S&P 500 constituents with full return histories ($q = 0.972$). Table 1 compares the three σ^2 estimators; the naive median produces a markedly inflated RMT count, motivating the use of the iterative trim throughout. Table 2 reports the three dimensionality estimates.

σ^2 method	σ^2	λ_+	RMT count
Bottom-half median	1.10×10^{-5}	4.36×10^{-5}	270
Iterative trim	4.13×10^{-5}	1.63×10^{-4}	149
MP density fit	4.01×10^{-5}	1.58×10^{-4}	152

Table 1: σ^2 estimator comparison on the 2018–2019 window.

Estimator	Value	Type
RMT (iterative σ^2)	149	integer / linear
Participation Ratio	10.24	continuous / linear
Levina–Bickel ($k = 10$)	29.48	continuous / nonlinear

Table 2: Dimensionality estimates on the 2018–2019 window.

The three estimators disagree by an order of magnitude, we think this is by design. RMT counts noise-distinguishable linear modes; PR measures spectral concentration; LB captures geometric manifold dimension. The gap LB − PR ≈ 20 quantifies the contribution of nonlinear curvature in

the return manifold that linear methods cannot see. PC1 alone explains 30.1% of total variance and has uniformly same-signed loadings, the textbook signature of a market mode.

3.2 Rolling Window Across 25 Years

Computing all three estimators on 252-day windows stepped quarterly across 2000–2024 produces 96 observations. The variance share of the top eigenvalue serves as a regime indicator. It exhibits three distinct spikes exceeding 0.50: late 2008 (Lehman), late 2011–mid 2012 (European sovereign debt crisis and U.S. credit downgrade), and 2020 (COVID). Notably, the 2011–12 spike was not a labeled regime in our analysis but emerges unprompted, providing a validation of the metric. PR and LB both compress during these windows; RMT moves less dramatically because $q > 1$ caps its magnitude through rank deficiency.

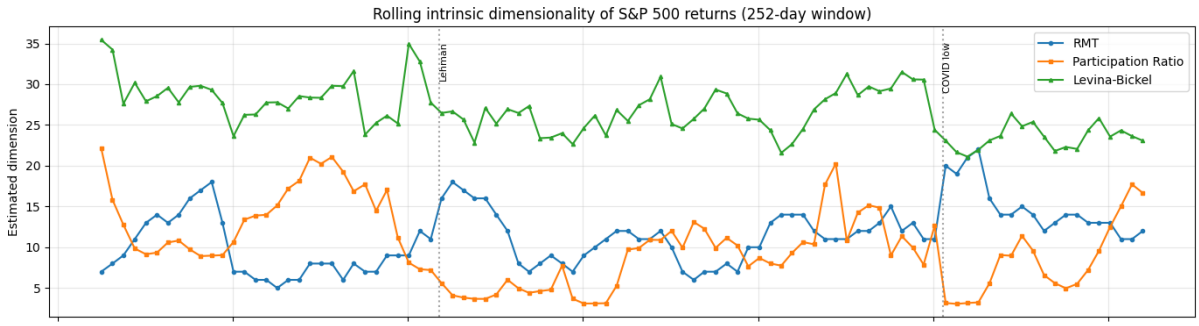


Figure 1: The two-tailed 95% estimator confidence intervals.

3.3 Robustness

Table 3 summarizes the two stress tests on the 2018–2019 window. Subsampling intervals are tight (CV under 3% for all three estimators), confirming that none of the estimates depend on which particular 80% of assets is drawn. Bootstrap intervals tell a more nuanced story: PR is well-centered on its point estimate, but RMT and LB are biased substantially downward. This is not a defect of the estimators but a consequence of moving-block resampling: duplicate rows reduce the effective rank of the bootstrap covariance, which caps RMT (rank-bound) and shrinks Levina–Bickel local distances toward zero. PR is invariant to this distortion because it is a scale- and rank-invariant ratio of spectral moments. We therefore use subsampling for RMT and LB confidence reporting and bootstrap for PR.

Estimator	Point	Sub. mean (CV%)	Boot. mean	Boot. 95% CI
RMT	149.00	114.92 (2.77%)	60.65	[44.0, 77.0]
PR	10.24	10.17 (2.80%)	10.33	[7.17, 14.62]
LB	29.48	28.75 (2.15%)	15.08	[10.09, 21.00]

Table 3: Robustness to asset subsampling ($B = 100$, 80% of assets) and moving-block bootstrap ($B = 200$, 20-day blocks).

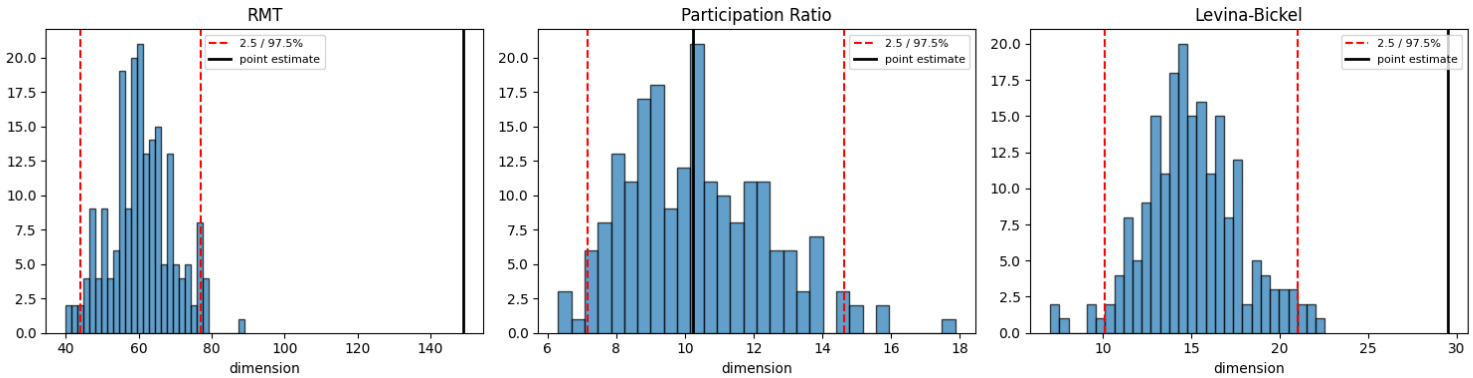


Figure 2: The two-tailed 95% estimator confidence intervals.

3.4 RMT-Cleaned Covariance and GMV Portfolios

Cleaning replaces 340 noise eigenvalues with their mean and preserves the 149 signal eigenvalues. The trace is preserved exactly. The condition number drops from 4.93×10^6 to 1.12×10^3, a 4400-fold improvement. Global minimum-variance portfolios constructed from the raw and cleaned covariances differ dramatically (Table 4): the raw GMV portfolio takes 10× more leverage and concentrates 85% of its weight in a single name, which is a noise-driven artifact. A cross-evaluation is informative: GMV(raw) evaluated under the cleaned covariance has variance 80× higher than GMV(cleaned)'s in-sample variance, evidence that the raw portfolio is overfit to noise.

Metric	GMV(raw)	GMV(cleaned)
Condition number of Σ	4.93×10^6	1.12×10^3
Gross leverage $\sum_i \lvert w_i \rvert$	68.30	6.48
Maximum $\lvert w_i \rvert$	0.85	0.06
Effective # names $1/\sum w_i^2$	0.05	7.25

Table 4: Global minimum-variance portfolios from raw and RMT-cleaned covariances on the 2018–2019 window.

3.5 Eigenportfolio Identification

Inspection of the top long and short positions in the L1-normalized eigenportfolios produces clean economic identities (Table 5). PC1 is the market mode; PC2 contrasts rate-sensitive REITs against oil and gas; PC3 separates defensives and retail from growth technology; PC4 isolates an energy basket; PC5 inverts PC4. Realized returns over the 2018–2019 window are consistent with these labels (PC2 returned +12.7% with a Sharpe ratio of 1.75) reflecting the historical outperformance of REITs against energy in that period. These Sharpe ratios are in-sample and serve as economic identification, not as out-of-sample strategy claims.

4 Discussion and Limitations

There are four items we should pay special attention to. First, the moving-block bootstrap biases RMT and LB downward through induced rank deficiency, as discussed above; subsampling is the

Component (var. share)	Long leg	Short leg
PC1 (30.1%)	— (uniform)	— (high-beta semis)
PC2 (4.5%)	REITs (WELL, VTR, AMT)	Oil & gas (APA, DVN, MRO)
PC3 (3.0%)	Defensives (HBI, CPB, JWN)	Growth tech (AMD, NVDA, NOW)
PC4 (2.6%)	Energy (MRO, DVN, APA)	Consumer cyclicals
PC5 (2.3%)	Cyclicals & airlines	Energy (PC4 inverted)

Table 5: Top long and short positions in the first five eigenportfolios.

more honest robustness instrument for these two estimators. Second, the eigenportfolio Sharpe ratios are computed in-sample and identify what each factor was over 2018–2019 but make no out-of-sample performance claim. [I'm not too sure about this next one JY] Third, raw PCA conflates the market factor with sector-level beta amplification, the high-beta semiconductor tilt of PC1 is a manifestation of this. A Fama–French market-orthogonalized analysis would isolate sector and style structure more cleanly in PC2–PC5, which is a natural extension. Finally, the dimensionality proposed by RMT is highly sensitive to the return window. As previously discussed, when $q > 1$, it's magnitude is capped via rank deficiency. Likewise, the λ_+ is dependent upon q. When $p >> n$, the number of statistically significant eigenvalues drops sharply. The focus of this work is centered on the S&P500 index, which is composed of approximately 500 unique assets at all times. As such, to avoid inflating the Marchenko-Pastur upper edge, rolling windows should contain longer returns series.

Author Contributions [placeholder for now, update as you please]

- **John van Brederode:** theoretical foundation, object-oriented library design, σ^2 estimators.
- **Jun Yen Lim:** data gathering, notebook implementation (single-window and rolling analysis, RMT cleaning, eigenportfolios),
- **Richard Sebek:** proposal authorship, literature review, presentation deck, Levina-Bickel background.

AI Use Case

- Jun Yen used AI to brainstorm possible solutions to coding errors and debug code.
- Jun Yen offloaded visualizations to AI as he believes front-end graphs are not the focus of this course.
- John van Brederode used AI to brainstorm class structure design and to debug errors.
- John van Brederode used AI to learn how to improve efficiency of existing code

References

[1] Laurent Laloux, Pierre Cizeau, Marc Potters, and Jean-Philippe Bouchaud. Random matrix theory and financial correlations. *Physical Review Letters*, 83(7):1467, 1999.

[2] Elizaveta Levina and Peter J. Bickel. Maximum likelihood estimation of intrinsic dimension. *Advances in Neural Information Processing Systems*, 17, 2004.

Intrinsic_Dimensionality_Consolidated

May 3, 2026

[32]:
```
# This cell ought** to contain everything necessary for the following code to␣
↪run. - Andy
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.linalg import eigh
from sklearn.neighbors import NearestNeighbors
from collections import Counter
from scipy.optimize import minimize_scalar


%matplotlib inline
np.set_printoptions(precision=4, suppress=True)
SEED = 6740

np.set_printoptions(precision=4, suppress=True)
pd.set_option("display.float_format", lambda x: f"{x:.4f}")
```

1 Data Preparation

[33]:
```
returns_full = pd.read_parquet("data_raw/sp500_returns_matrix.parquet")
returns_full.index = pd.to_datetime(returns_full.index)
returns_full = returns_full.sort_index()

constituent_history = pd.read_parquet("data_raw/sp500_constituent_history.
↪parquet")
constituent_history["date"] = pd.to_datetime(constituent_history["date"])
constituent_history = constituent_history.sort_values("date").
↪reset_index(drop=True)

permno_ticker_map = pd.read_parquet("data_raw/permno_ticker_map.parquet")

print(f"returns_full:        {returns_full.shape}, {returns_full.index.min().
↪date()} \u2192 {returns_full.index.max().date()}")
print(f"constituent_history: {constituent_history.shape},␣
↪{len(constituent_history)} membership snapshots")
```

```python
print(f"permno_ticker_map:   {permno_ticker_map.shape}")
```

```
returns_full:        (6289, 1322), 2000-01-03 → 2024-12-31
constituent_history: (2213, 2), 2213 membership snapshots
permno_ticker_map:   (5456, 5)
```

[34]:
```python
def members_as_of(date, history=constituent_history):
    """
    Purpose: Pulls the tickers S&P members 'as of' a given date (binary search␣
 ↪on snapshot dates). This prevents survivorship bias since the S&P prunes the␣
 ↪'losers.'

    Inputs: Date and constituency

    Outputs: A list of stock tickers
    """
    target = pd.to_datetime(date)
    idx = history["date"].searchsorted(target, side="right") - 1
    if idx < 0:
        raise ValueError(f"No constituent record on or before {date}")
    return set(history.iloc[idx]["tickers"].split(","))


# Returns_full column -> base ticker (strip the _PERMNO suffix used only when␣
 ↪CRSP reused a ticker across companies). Cached as a Series.
_base_ticker = pd.Series(
    [c.split("_")[0] for c in returns_full.columns],
    index=returns_full.columns,
)


def members_to_columns(member_tickers):
    """
    Purpose: Builds a dataframe with the returns (over a predetermined window)␣
 ↪of the companies in the member_tickers list.

    Inputs: A list of tickers

    Outputs: returns_full columns whose base ticker is in the membership set.
    """
    return returns_full.columns[_base_ticker.isin(member_tickers)]


# Lambda function: Maps column label to (ticker, company name) using the␣
 ↪permno_ticker_map
_pt_latest = permno_ticker_map.sort_values("namedt").groupby("permno",␣
 ↪as_index=False).last()[["permno", "ticker", "comnam"]]
```

```
_ticker_freq = Counter(_pt_latest["ticker"])
_pt_latest["col_label"] = _pt_latest.apply(
    lambda r: f"{r['ticker']}_{r['permno']}" if _ticker_freq[r["ticker"]] > 1␣
 ↪else r["ticker"],
    axis=1,
)
col_to_company = dict(zip(_pt_latest["col_label"], _pt_latest["comnam"]))

print("Testing Data Processing:")
print(f"  Example \u2014 members_as_of('2019-12-31'): ␣
 ↪{len(members_as_of('2019-12-31'))} tickers")
print(f"  Example \u2014 col_to_company['ORCL']:       {col_to_company.
 ↪get('ORCL')}")
print(f"  Example \u2014 col_to_company['JAVA_10078']: {col_to_company.
 ↪get('JAVA_10078')}")

print(f"returns_full:        {returns_full.shape}, {returns_full.index.min().
 ↪date()} → {returns_full.index.max().date()}")
print(f"constituent_history: {constituent_history.shape}")
print(f"permno_ticker_map:   {permno_ticker_map.shape}")
```

```
Testing Data Processing:
  Example - members_as_of('2019-12-31'):  505 tickers
  Example - col_to_company['ORCL']:       ORACLE CORP
  Example - col_to_company['JAVA_10078']: SUN MICROSYSTEMS INC
returns_full:        (6289, 1322), 2000-01-03 → 2024-12-31
constituent_history: (2213, 2)
permno_ticker_map:   (5456, 5)
```

[35]:
```
# Single-window analysis. 2-year pre-COVID baseline, which keeps q = p/n < 1␣
 ↪and avoids the 2020 vol spike from contaminating the noise-bulk estimate.
AS_OF_DATE = "2019-12-31"
WINDOW_START = "2018-01-01"
WINDOW_END = "2019-12-31"

# Real S&P 500 membership at the as-of date
members = members_as_of(AS_OF_DATE)
member_cols = members_to_columns(members)

# Slice to window and member columns
returns_win = returns_full.loc[WINDOW_START:WINDOW_END, member_cols]

# Drop assets with any missing return in the window (no imputation)
returns_win = returns_win.dropna(axis=1, how="any")

# Cast from pandas nullable Float64 to NumPy float64 for downstream math
X_raw = returns_win.to_numpy(dtype=float)
```

```
n, p = X_raw.shape
q = p / n

print(f"As-of date:                 {AS_OF_DATE}")
print(f"S&P members at as-of:       {len(members)}")
print(f"Members with any data:      {len(member_cols)}")
print(f"Members with full window:   {p}")
print(f"Window:                     {WINDOW_START} → {WINDOW_END}")
print(f"  n (days):                 {n}")
print(f"  p (assets):               {p}")
print(f"  q = p/n:                  {q:.3f}")
```

```
As-of date:                 2019-12-31
S&P members at as-of:       505
Members with any data:      589
Members with full window:   489
Window:                     2018-01-01 → 2019-12-31
  n (days):                 503
  p (assets):               489
  q = p/n:                  0.972
```

[36]:
```
# Demean (column-wise) and build covariance via X.T @ X
X = X_raw - X_raw.mean(axis=0, keepdims=True)
cov = (X.T @ X) / (n - 1)

# Eigendecomposition. eigh returns ascending evals, so we need to flip the␣
↪indexing to descending.
evals, evecs = eigh(cov)
evals = evals[::-1]
evecs = evecs[:, ::-1]

print(f"Covariance shape:     {cov.shape}")
print(f"Top 5 eigenvalues:    {evals[:5]}")
print(f"Total variance (\u03a3\u03bb): {evals.sum():.6e}")
print(f"Top eig share:        {evals[0] / evals.sum():.1%}")
```

```
Covariance shape:     (489, 489)
Top 5 eigenvalues:    [0.0461 0.0068 0.0046 0.0039 0.0035]
Total variance (Σ λ): 1.529194e-01
Top eig share:        30.1%
```

2 Method 1 - Marchenko-Pastur (RMT)

[37]:
```
def mp_density(lam, sigma2_, q_):
    """
    Purpose: Marchenko-Pastur continuous density at points `lam`.
```

```
    Inputs: Eigenvalue, noise estimate, q_ = p/n

    Outputs: Marchenko-Pastur Density
    """

    lam_p = sigma2_ * (1 + np.sqrt(q_)) ** 2
    lam_m = sigma2_ * (1 - np.sqrt(q_)) ** 2
    out = np.zeros_like(lam, dtype=float)
    inside = (lam > lam_m) & (lam < lam_p)
    out[inside] = np.sqrt((lam_p - lam[inside]) * (lam[inside] - lam_m)) / (2 *␣
↪np.pi * sigma2_ * q_ * lam[inside])
    return out


def iter_sigma2(evals_, q_, max_iter=100, tol=1e-12):
    """
    Purpose: Iteratively trim eigenvalues above the implied MP edge;  ² is the␣
↪mean of what remains

    Filters to strictly positive eigenvalues so rank-deficiency artifacts (when␣
↪the
    sample cov is rank < p - happens with q > 1 OR with bootstrap-resampled data
    that has duplicate rows) don't drag  ² toward zero and inflate the signal␣
↪count.

    Inputs: Eigenvalues, q_ = p/n, max_iterations, convergence criteria

    Outputs: Noise estimate,  ²
    """
    positive = evals_[evals_ > 1e-14 * max(evals_.max(), 1e-30)]
    if len(positive) == 0:
        return float(max(evals_.max(), 1e-30))
    sigma2_ = positive.mean()
    for _ in range(max_iter):
        edge = sigma2_ * (1 + np.sqrt(q_)) ** 2
        keep = positive <= edge
        if keep.sum() == 0:
            break
        new_sigma2 = positive[keep].mean()
        if abs(new_sigma2 - sigma2_) < tol * sigma2_:
            sigma2_ = new_sigma2
            break
        sigma2_ = new_sigma2
    return sigma2_


def fit_sigma2(evals_, q_, bulk_quantile=0.9, n_bins=50):
```

```
    """
    Purpose: Choose ² to minimize L² distance between empirical bulk histogram␣
↪and MP density.

    Filters to strictly positive eigenvalues so rank-deficiency artifacts (when␣
↪the
    sample cov is rank < p - happens with q > 1 OR with bootstrap-resampled data
    that has duplicate rows) don't drag ² toward zero and inflate the signal␣
↪count.

    Inputs: Eigenvalues, q_ = p/n, quantile, number of histogram bins

    Outputs: Noise estimate, ²
    """
    cutoff = np.quantile(evals_, bulk_quantile)
    bulk = evals_[evals_ <= cutoff]
    hist, edges = np.histogram(bulk, bins=n_bins, density=True)
    centers = 0.5 * (edges[1:] + edges[:-1])
    s0 = evals_.mean()
    res = minimize_scalar(
        lambda s2: np.sum((hist - mp_density(centers, s2, q_)) ** 2),
        bounds=(s0 * 0.05, s0 * 5),
        method="bounded",
    )
    return res.x


def rmt_count(s2, evs, q_):
    return int((evs > s2 * (1 + np.sqrt(q_)) ** 2).sum())


# Compare the three ² estimators on the single-window spectrum
sigma2_median = np.median(evals[p // 2 :]) # median of the lower half method
sigma2_iter = iter_sigma2(evals, q)
sigma2_fit = fit_sigma2(evals, q)

print(f"{'method':<22}  {' ²':>14}  {' +':>14}  {'#signal':>8}")
for name, s2 in [
    ("bottom-half median", sigma2_median),
    ("iterative trim", sigma2_iter),
    ("MP density fit", sigma2_fit),
]:
    edge = s2 * (1 + np.sqrt(q)) ** 2
    print(f"{name:<22}  {s2:>14.4e}  {edge:>14.4e}  {rmt_count(s2, evals, q):
↪>8d}")
```

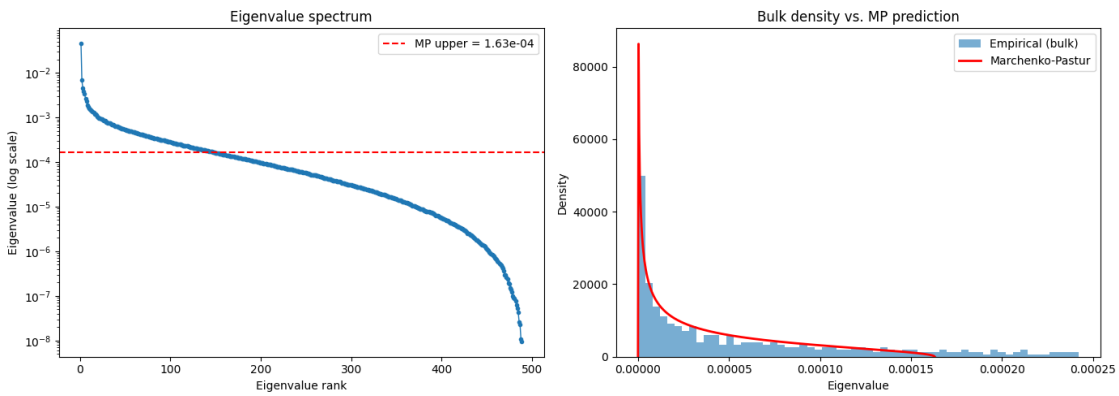
```
# Iterative trim will be the standard approach going forward, or as my␣
↪teammates (probably just John) like to say "canonical."
sigma2 = sigma2_iter
mp_upper = sigma2 * (1 + np.sqrt(q)) ** 2
mp_lower = sigma2 * (1 - np.sqrt(q)) ** 2 if q < 1 else 0.0
rmt_dim = rmt_count(sigma2, evals, q)
print(f"\nCanonical RMT dimension (iterative): {rmt_dim}")
print(f"Top {min(rmt_dim, 10)} signal eigenvalues: {evals[:min(rmt_dim, 10)]}")
```

```
method                                  ²                +   #signal
bottom-half median          1.1042e-05      4.3553e-05       270
iterative trim              4.1321e-05      1.6298e-04       149
MP density fit              4.0122e-05      1.5825e-04       152

Canonical RMT dimension (iterative): 149
Top 10 signal eigenvalues: [0.0461 0.0068 0.0046 0.0039 0.0035 0.0027 0.0024
0.0019 0.0017 0.0016]
```

[38]:

```
"""
tl;dr if you are above the red lines in either plot, you aren't what MP says␣
↪you should be, and therefore, statistically significant.
"""

fig, axes = plt.subplots(1, 2, figsize=(14, 5))


# Left: full spectrum (log scale) with MP threshold
axes[0].plot(np.arange(1, len(evals) + 1), evals, marker=".", linestyle="-",␣
↪linewidth=1)
axes[0].axhline(mp_upper, color="red", linestyle="--", label=f"MP upper =␣
↪{mp_upper:.2e}")
axes[0].set_xlabel("Eigenvalue rank")
axes[0].set_ylabel("Eigenvalue (log scale)")
axes[0].set_yscale("log")
axes[0].set_title("Eigenvalue spectrum")
axes[0].legend()

# Right: empirical density of the bulk vs. theoretical MP density
bulk = evals[evals < mp_upper * 1.5]
axes[1].hist(bulk, bins=60, density=True, alpha=0.6, label="Empirical (bulk)")

lam = np.linspace(max(mp_lower, 1e-12), mp_upper, 500)
mp_density = np.sqrt(np.maximum((mp_upper - lam) * (lam - mp_lower), 0)) / (2 *␣
↪np.pi * sigma2 * q * lam)
axes[1].plot(lam, mp_density, color="red", linewidth=2,␣
↪label="Marchenko-Pastur")
axes[1].set_xlabel("Eigenvalue")
```

```
axes[1].set_ylabel("Density")
axes[1].set_title("Bulk density vs. MP prediction")
axes[1].legend()

plt.tight_layout()
plt.show()
```



3 Method 2 - Participation Ratio (PR)

[39]:
```
# This will be functionalized later
pr_dim = (evals.sum() ** 2) / (evals**2).sum()
print(f"Participation ratio dimension:  {pr_dim:.2f}")
print(f"  (variance spread is equivalent to ~{pr_dim:.0f} equally-weighted␣
↪modes)")
```

```
Participation ratio dimension:  10.24
  (variance spread is equivalent to ~10 equally-weighted modes)
```

4 Method 3 - Levina-Bickel (LB)

[40]:
```
# This will be functionalized later
K = 10  # neighbor count

nbrs = NearestNeighbors(n_neighbors=K + 1).fit(X)
distances, _ = nbrs.kneighbors(X)
# distances[:, 0] is the self-distance (= 0); distances[:, 1:K+1] are the K␣
↪nearest neighbors

T_k = distances[:, K]  # shape (n,) - distance to k-th NN
T_j = distances[:, 1:K]  # shape (n, K-1) - distances to NN 1..K-1
```

```
eps = 1e-12  # guard against zero distances
log_ratios = np.log((T_k[:, None] + eps) / (T_j + eps))
local_dims = (K - 1) / log_ratios.sum(axis=1)
local_dims = local_dims[np.isfinite(local_dims)]

lb_dim = local_dims.mean()

print(f"Levina-Bickel dimension (k={K}):  {lb_dim:.2f}")
print(f"  Averaged over {len(local_dims)} valid sample points")
```

```
Levina-Bickel dimension (k=10):  29.48
  Averaged over 503 valid sample points
```

5 Dimensionality Estimate Comparisons

[41]:
```
summary = pd.DataFrame(
    {
        "estimator": ["RMT (MP upper edge)", "Participation Ratio",␣
 ↪"Levina-Bickel"],
        "dimension": [rmt_dim, round(pr_dim, 2), round(lb_dim, 2)],
        "type": ["integer / linear", "continuous / linear", "continuous /␣
 ↪nonlinear"],
    }
)
print(f"Window: {WINDOW_START} to {WINDOW_END}  (n={n}, p={p}, q={q:.3f})\n")
print(summary.to_string(index=False))
```

```
Window: 2018-01-01 to 2019-12-31  (n=503, p=489, q=0.972)

          estimator  dimension                   type
RMT (MP upper edge)   149.0000       integer / linear
Participation Ratio    10.2400    continuous / linear
      Levina-Bickel    29.4800 continuous / nonlinear
```

[42]:
```
pc1 = evecs[:, 0]
pc1_df = pd.DataFrame(
    {
        "ticker": returns_win.columns,
        "company": [col_to_company.get(c, "?") for c in returns_win.columns],
        "weight": pc1,
    }
)
pc1_df["abs_weight"] = pc1_df["weight"].abs()

print("Top 10 PC1 loadings by absolute weight:")
print(pc1_df.sort_values("abs_weight", ascending=False).head(10).
 ↪to_string(index=False))
```

```
print()
print(f"PC1 sign distribution:  {(pc1 > 0).sum()} positive, {(pc1 < 0).sum()}␣
↪negative")
print(f"  (mostly same-sign \u2192 broad market mode; mixed \u2192 sector or␣
↪style factor)")
```

```
Top 10 PC1 loadings by absolute weight:
   ticker                    company  weight  abs_weight
      AMD ADVANCED MICRO DEVICES INC -0.0953      0.0953
     NVDA                NVIDIA CORP -0.0919      0.0919
      URI         UNITED RENTALS INC -0.0893      0.0893
       MU      MICRON TECHNOLOGY INC -0.0887      0.0887
     IPGP       I P G PHOTONICS CORP -0.0816      0.0816
      WDC       WESTERN DIGITAL CORP -0.0775      0.0775
FCX_81774       FREEPORT MCMORAN INC -0.0763      0.0763
     LRCX              LAM RESH CORP -0.0751      0.0751
     SIVB      S V B FINANCIAL GROUP -0.0749      0.0749
     AMAT      APPLIED MATERIALS INC -0.0743      0.0743

PC1 sign distribution:  0 positive, 489 negative
  (mostly same-sign → broad market mode; mixed → sector or style factor)
```

[43]:
```
WIN_DAYS = 252  # 1 trading year
STEP_DAYS = 63  # Roughly 1 trading quarter
K_ROLL = 10

dates = returns_full.index
records = []

for start_idx in range(0, len(dates) - WIN_DAYS + 1, STEP_DAYS):
    end_idx = start_idx + WIN_DAYS
    end_date = dates[end_idx - 1]

    # Per-window membership: S&P constituents at the window's end date
    members_w = members_as_of(end_date)
    cols_w = members_to_columns(members_w)

    win = returns_full.iloc[start_idx:end_idx][cols_w].dropna(axis=1, how="any")
    n_w, p_w = win.shape
    if p_w < 20:
        continue

    Xw = win.to_numpy(dtype=float)
    Xw = Xw - Xw.mean(axis=0, keepdims=True)
    cov_w = (Xw.T @ Xw) / (n_w - 1)
    evals_w = eigh(cov_w, eigvals_only=True)[::-1]
    q_w = p_w / n_w
```

```
    # RMT with iterative  ² (the same as the single-window cell)
    sigma2_w = iter_sigma2(evals_w, q_w)
    mp_upper_w = sigma2_w * (1 + np.sqrt(q_w)) ** 2
    rmt_w = int(np.sum(evals_w > mp_upper_w))

    # Participation ratio
    pr_w = (evals_w.sum() ** 2) / (evals_w**2).sum()

    # Levina-Bickel (vectorized inside the window)
    nbrs_w = NearestNeighbors(n_neighbors=K_ROLL + 1).fit(Xw)
    d_w, _ = nbrs_w.kneighbors(Xw)
    Tk_w = d_w[:, K_ROLL]
    Tj_w = d_w[:, 1:K_ROLL]
    lr_w = np.log((Tk_w[:, None] + 1e-12) / (Tj_w + 1e-12))
    ld_w = (K_ROLL - 1) / lr_w.sum(axis=1)
    lb_w = ld_w[np.isfinite(ld_w)].mean()

    records.append(
        {
            "window_end": end_date,
            "n_assets": p_w,
            "rmt": rmt_w,
            "pr": pr_w,
            "lb": lb_w,
            "top_eig_share": evals_w[0] / evals_w.sum(),
        }
    )

rolling = pd.DataFrame(records).set_index("window_end")
print(f"Computed {len(rolling)} rolling windows ({WIN_DAYS}-day, step␣
 ↪{STEP_DAYS}d)")
rolling.head()
```

```
Computed 96 rolling windows (252-day, step 63d)
```

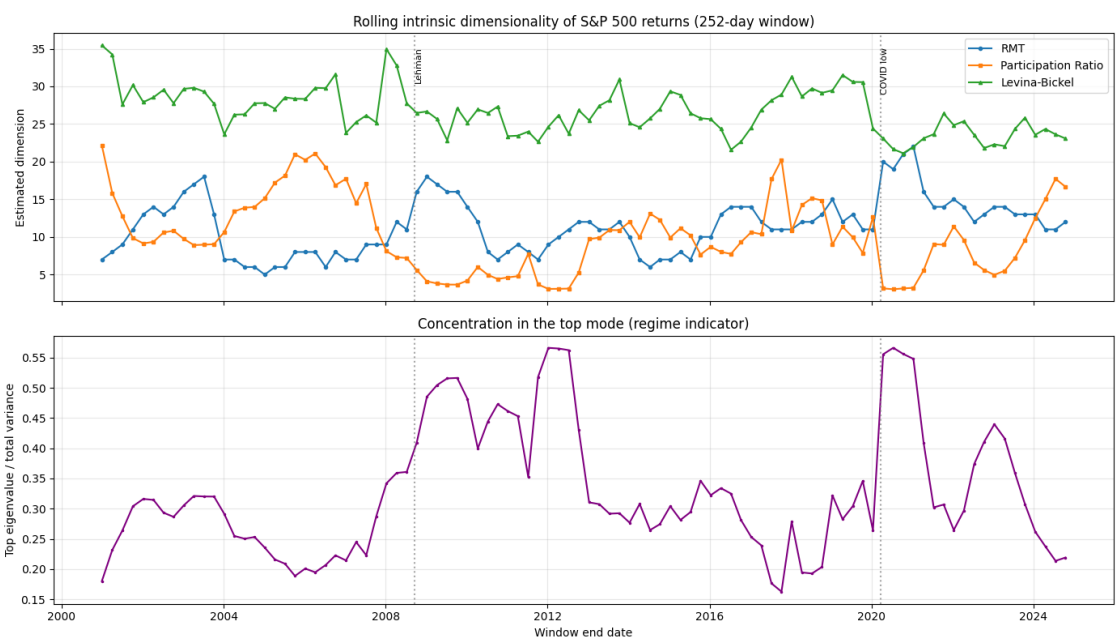
[43]:

window_end	n_assets	rmt	pr	lb	top_eig_share
2000-12-29	482	7	22.1567	35.4462	0.1800
2001-04-02	482	8	15.8339	34.2472	0.2316
2001-07-02	482	9	12.7642	27.6002	0.2637
2001-10-05	478	11	9.8446	30.1760	0.3039
2002-01-07	481	13	9.1379	27.9025	0.3161

[44]:

```
fig, axes = plt.subplots(2, 1, figsize=(14, 8), sharex=True)

axes[0].plot(rolling.index, rolling["rmt"], label="RMT", marker="o",␣
 ↪markersize=3)
```

```
axes[0].plot(rolling.index, rolling["pr"], label="Participation Ratio",␣
 ↪marker="s", markersize=3)
axes[0].plot(rolling.index, rolling["lb"], label="Levina-Bickel", marker="^",␣
 ↪markersize=3)
axes[0].set_ylabel("Estimated dimension")
axes[0].set_title("Rolling intrinsic dimensionality of S&P 500 returns (252-day␣
 ↪window)")
axes[0].legend()
axes[0].grid(alpha=0.3)

axes[1].plot(rolling.index, rolling["top_eig_share"], color="purple", marker=".
 ↪", markersize=3)
axes[1].set_ylabel("Top eigenvalue / total variance")
axes[1].set_xlabel("Window end date")
axes[1].set_title("Concentration in the top mode (regime indicator)")
axes[1].grid(alpha=0.3)

for date_str, label in [("2008-09-15", "Lehman"), ("2020-03-23", "COVID low")]:
    d = pd.Timestamp(date_str)
    if rolling.index.min() <= d <= rolling.index.max():
        for ax in axes:
            ax.axvline(d, color="gray", linestyle=":", alpha=0.7)
        axes[0].text(d, axes[0].get_ylim()[1] * 0.95, label, rotation=90,␣
 ↪va="top", fontsize=8)

plt.tight_layout()
plt.show()
```



6 Evaluation - Random Asset Subsampling

[45]:
```
B_SUB = 100
SUB_FRAC = 0.80
rng_sub = np.random.default_rng(SEED)

p_sub = int(p * SUB_FRAC)
sub_results = {"rmt": [], "pr": [], "lb": []}

for _ in range(B_SUB):
    cols = rng_sub.choice(p, size=p_sub, replace=False)
    Xs = X[:, cols]
    cov_s = (Xs.T @ Xs) / (n - 1)
    es = eigh(cov_s, eigvals_only=True)[::-1]
    qs = p_sub / n

    # RMT (iterative  ²)
    s2_s = iter_sigma2(es, qs)
    edge_s = s2_s * (1 + np.sqrt(qs)) ** 2
    sub_results["rmt"].append(int((es > edge_s).sum()))

    # Participation ratio
    sub_results["pr"].append((es.sum() ** 2) / (es**2).sum())

    # Levina-Bickel
    nbrs_s = NearestNeighbors(n_neighbors=K + 1).fit(Xs)
    d_s, _ = nbrs_s.kneighbors(Xs)
    Tk_s = d_s[:, K]
    Tj_s = d_s[:, 1:K]
    lr_s = np.log((Tk_s[:, None] + 1e-12) / (Tj_s + 1e-12))
    ld_s = (K - 1) / lr_s.sum(axis=1)
    sub_results["lb"].append(ld_s[np.isfinite(ld_s)].mean())

stress = pd.DataFrame(
    {
        "estimator": ["RMT", "Participation Ratio", "Levina-Bickel"],
        "full_window": [rmt_dim, round(pr_dim, 2), round(lb_dim, 2)],
        "sub_mean": [np.mean(sub_results[k]) for k in ["rmt", "pr", "lb"]],
        "sub_std": [np.std(sub_results[k]) for k in ["rmt", "pr", "lb"]],
        "sub_cv_pct": [np.std(sub_results[k]) / np.mean(sub_results[k]) * 100␣
 ↪for k in ["rmt", "pr", "lb"]],
    }
)
print(f"80% asset subsampling, B={B_SUB} replicates")
print(stress.to_string(index=False, float_format=lambda x: f"{x:.2f}"))
```

```
80% asset subsampling, B=100 replicates
          estimator  full_window  sub_mean  sub_std  sub_cv_pct
                RMT       149.00    114.92     3.19        2.77
Participation Ratio        10.24     10.17     0.29        2.80
```

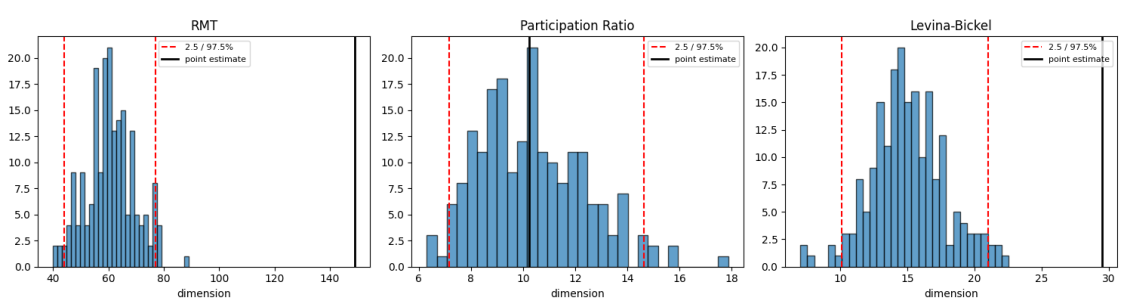
```
Levina-Bickel        29.48     28.75     0.62        2.15
```

7 Evaluation - Moving-Block Bootstrap CI

[46]:
```
# Okay, the code wizards in this group love to torture me with their evil␣
↪heiroglyphs of despair - Andy
def estimators_from_X(Xb, K_=10):
    """
    Purpose: Takes returns matrix and converts it into all three dimensionality␣
↪estimates

    Inputs: data matrix, # of nearest neighbors

    Outputs: # of eigenvalues above the Marchenko-Pastur upper edge,␣
↪participation ratio, Levina-Bickel estimate
    """
    nb_, pb_ = Xb.shape
    Xb = Xb - Xb.mean(axis=0, keepdims=True)
    cov_b = (Xb.T @ Xb) / (nb_ - 1)
    eb = eigh(cov_b, eigvals_only=True)[::-1]
    qb = pb_ / nb_
    s2_b = iter_sigma2(eb, qb)
    edge_b = s2_b * (1 + np.sqrt(qb)) ** 2
    rmt_b = int((eb > edge_b).sum())
    pr_b = (eb.sum() ** 2) / (eb**2).sum()
    nbrs_b = NearestNeighbors(n_neighbors=K_ + 1).fit(Xb)
    db_, _ = nbrs_b.kneighbors(Xb)
    Tk_b, Tj_b = db_[:, K_], db_[:, 1:K_]
    lr_b = np.log((Tk_b[:, None] + 1e-12) / (Tj_b + 1e-12))
    ld_b = (K_ - 1) / lr_b.sum(axis=1)
    lb_b = ld_b[np.isfinite(ld_b)].mean()
    return rmt_b, pr_b, lb_b


B_BOOT = 200
BLOCK = 20  # approx. one trading month
rng_boot = np.random.default_rng(SEED)

n_blocks = int(np.ceil(n / BLOCK))
boot = {"rmt": [], "pr": [], "lb": []}
for _ in range(B_BOOT):
    starts = rng_boot.integers(0, n - BLOCK + 1, size=n_blocks)
    idx = np.concatenate([np.arange(s, s + BLOCK) for s in starts])[:n]
    Xb = X_raw[idx]
    r_, p_, l_ = estimators_from_X(Xb, K_=K)
    boot["rmt"].append(r_)
```

```
    boot["pr"].append(p_)
    boot["lb"].append(l_)

ci_df = pd.DataFrame(
    {
        "estimator": ["RMT", "Participation Ratio", "Levina-Bickel"],
        "point": [rmt_dim, round(pr_dim, 2), round(lb_dim, 2)],
        "boot_mean": [np.mean(boot[k]) for k in ["rmt", "pr", "lb"]],
        "ci_2.5": [np.percentile(boot[k], 2.5) for k in ["rmt", "pr", "lb"]],
        "ci_97.5": [np.percentile(boot[k], 97.5) for k in ["rmt", "pr", "lb"]],
        "width": [np.percentile(boot[k], 97.5) - np.percentile(boot[k], 2.5)␣
 ↪for k in ["rmt", "pr", "lb"]],
    }
)
print(f"Moving-block bootstrap, B={B_BOOT}, block_len={BLOCK} days")
print(ci_df.to_string(index=False, float_format=lambda x: f"{x:.2f}"))

fig, axes = plt.subplots(1, 3, figsize=(15, 4))
points = {"rmt": rmt_dim, "pr": pr_dim, "lb": lb_dim}
names = {"rmt": "RMT", "pr": "Participation Ratio", "lb": "Levina-Bickel"}
for ax, key in zip(axes, ["rmt", "pr", "lb"]):
    ax.hist(boot[key], bins=30, alpha=0.7, edgecolor="k")
    ax.axvline(np.percentile(boot[key], 2.5), color="red", linestyle="--",␣
 ↪label="2.5 / 97.5%")
    ax.axvline(np.percentile(boot[key], 97.5), color="red", linestyle="--")
    ax.axvline(points[key], color="black", linewidth=2, label="point estimate")
    ax.set_title(names[key])
    ax.set_xlabel("dimension")
    ax.legend(fontsize=8)
plt.tight_layout()
plt.show()
```

```
Moving-block bootstrap, B=200, block_len=20 days
          estimator  point  boot_mean  ci_2.5  ci_97.5  width
                RMT 149.00      60.65   44.00    77.03  33.03
Participation Ratio  10.24      10.33    7.17    14.62   7.45
      Levina-Bickel  29.48      15.07   10.08    20.99  10.91
```



[ ]:

```

```

8 Formalizing Methodologies

[47]:

```
# rmt_dim_from_evals is a slight change-up from rmt_count, but it does␣
↪effectively the same thing! We had some hiccups with later functions, and
# for convencience sake, we decided to just define a second function instead of␣
↪reconciling all of the later functions.

# Also, this is just functionalizing the methods that have already been␣
↪explained, so I won't leave specific commentary on the purpose + I/O of them.

def rmt_dim_from_evals(evs, q_):
    s2 = iter_sigma2(evs, q_)
    edge = s2 * (1 + np.sqrt(q_)) ** 2
    return int((evs > edge).sum()), s2, edge

def participation_ratio(evs):
    return float((evs.sum() ** 2) / (evs**2).sum())

def levina_bickel(X_, k=10):
    nbrs = NearestNeighbors(n_neighbors=k + 1).fit(X_)
    d, _ = nbrs.kneighbors(X_)
    Tk = d[:, k]
    Tj = d[:, 1:k]
    lr = np.log((Tk[:, None] + 1e-12) / (Tj + 1e-12))
    ld = (k - 1) / lr.sum(axis=1)
    return float(ld[np.isfinite(ld)].mean())


rmt_dim, sigma2, mp_upper = rmt_dim_from_evals(evals, q)
pr_dim = participation_ratio(evals)
lb_dim = levina_bickel(X, k=10)

print(f"RMT (iterative ²):   {rmt_dim}    (²={sigma2:.4e},  +={mp_upper:.4e})")
print(f"Participation Ratio:  {pr_dim:.2f}")
print(f"Levina-Bickel (k=10): {lb_dim:.2f}")
```

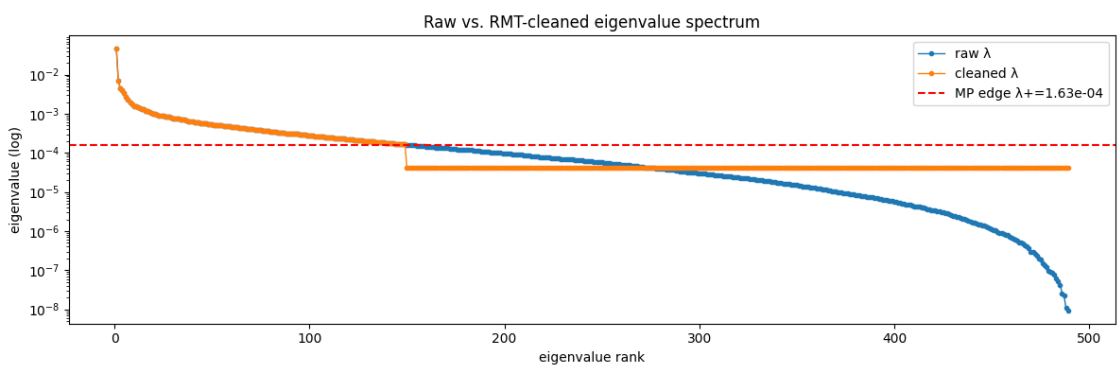
```
RMT (iterative ²):   149    (²=4.1321e-05,  +=1.6298e-04)
Participation Ratio:  10.24
Levina-Bickel (k=10): 29.48
```

9 RMT - Covariance Matrix (Cleaned Covariance)

[48]:
```
def rmt_clean_covariance(evals_, evecs_, mp_upper_):
    """
    Purpose: Replace noise eigenvalues with their mean, reconstruct symmetric␣
 ↪PSD cov. Makes a "clean" or "denoised" covariance matrix for asset returns :)

    Uses only positive noise eigenvalues to estimate the noise mean, and␣
 ↪handles the
    q > 1 case where some noise eigenvalues are numerically zero/negative due to
    rank deficiency. The fill value floors all noise eigenvalues at a strictly
    positive number, guaranteeing the reconstructed cov is positive definite.

    Inputs: eigenvalues, eigenvectors, the MP upper edge

    Outputs: cleaned cov matrix, denoised eigenvalues
    """

    e = evals_.copy()
    signal_mask = e > mp_upper_
    noise_mask = ~signal_mask
    if noise_mask.sum() > 0:
        positive_noise = e[noise_mask & (e > 0)]
        if len(positive_noise) > 0:
            noise_mean = positive_noise.mean()
        elif signal_mask.sum() > 0:
            noise_mean = e[signal_mask].min()
        else:
            noise_mean = max(mp_upper_, 1e-15)
        e[noise_mask] = noise_mean # Tentative research tells me that this is a␣
 ↪financial adaptation of Rotationally Invariant Estimators (RIEs). Neat!
    cleaned = evecs_ @ np.diag(e) @ evecs_.T
    cleaned = (cleaned + cleaned.T) / 2.0
    return cleaned, e


cov_cleaned, evals_cleaned = rmt_clean_covariance(evals, evecs, mp_upper)

cond_raw = evals[0] / evals[-1] if evals[-1] > 0 else np.inf
cond_cleaned = evals_cleaned.max() / evals_cleaned.min()
trace_raw = evals.sum()
trace_cleaned = evals_cleaned.sum()

print(f"Signal eigs kept:                    {(evals > mp_upper).sum()} of {p}")
print(f"Noise eigs replaced with mean:       {evals[(evals <= mp_upper) & (evals␣
 ↪> 0)].mean():.4e}")
print(f"Trace (raw):                         {trace_raw:.6e}")
```

```
print(f"Trace (cleaned):                    {trace_cleaned:.6e}    (preserved by␣
 ↪construction)")
print(f"Condition number (raw):             {cond_raw:.2e}")
print(f"Condition number (cleaned):         {cond_cleaned:.2e}    (~{cond_raw /␣
 ↪cond_cleaned:.0f}× better)")

fig, ax = plt.subplots(figsize=(12, 4))
rank = np.arange(1, p + 1)
ax.plot(rank, evals, label="raw  ", marker=".", linewidth=1)
ax.plot(rank, evals_cleaned, label="cleaned  ", marker=".", linewidth=1)
ax.axhline(mp_upper, color="red", linestyle="--", label=f"MP edge  +={mp_upper:.
 ↪2e}")
ax.set_yscale("log")
ax.set_xlabel("eigenvalue rank")
ax.set_ylabel("eigenvalue (log)")
ax.set_title("Raw vs. RMT-cleaned eigenvalue spectrum")
ax.legend()
plt.tight_layout()
plt.show()
```

```
Signal eigs kept:                   149 of 489
Noise eigs replaced with mean:      4.1321e-05
Trace (raw):                        1.529194e-01
Trace (cleaned):                    1.529194e-01    (preserved by construction)
Condition number (raw):             4.93e+06
Condition number (cleaned):         1.12e+03    (~4418× better)
```



10 Global Minimum-Variance Portfolios (Full vs. Cleaned Covariance Matrices)

[49]:
```
def gmv_weights(cov_):
    """
    Purpose: Get the asset weights needed to build the GMV portfolio

    Inputs: covariance matrix

    Outputs: portfolio weights of the Global Minimum-Variance Portfolio␣
 ↪(collection of assets that produces the last returns variance on the␣
 ↪efficient frontier)
    """
    ones = np.ones(cov_.shape[0])
    inv_cov_ones = np.linalg.solve(cov_, ones)
    return inv_cov_ones / (ones @ inv_cov_ones)


w_raw = gmv_weights(cov)
w_cleaned = gmv_weights(cov_cleaned)


def portfolio_metrics(w, label, cov_self, cov_other, other_label):
    """
    Purpose: Computes a set of diagnostic statistics for a given portfolio␣
 ↪weight vector under two different covariance matricies

    Inputs: weights, portfolio name (label), cov matrix to make the weights,␣
 ↪other cov matrix, other portfolio name (label)

    Outputs: portfolio weights of the Global Minimum-Variance Portfolio␣
 ↪(collection of assets that produces the last returns variance on the␣
 ↪efficient frontier)
    """
    return {
        "portfolio": label,
        f"variance under {label} cov": float(w @ cov_self @ w),
        f"variance under {other_label} cov": float(w @ cov_other @ w),
        "gross leverage Σ|w|": float(np.abs(w).sum()),
        "max |w|": float(np.abs(w).max()),
        "effective # names 1/Σw²": float(1.0 / (w**2).sum()),
        "# negative weights": int((w < 0).sum()),
    }


m_raw = portfolio_metrics(w_raw, "raw", cov, cov_cleaned, "cleaned")
```

```python
m_cleaned = portfolio_metrics(w_cleaned, "cleaned", cov_cleaned, cov, "raw")

# Pretty-print side by side
keys = list(m_raw.keys())
for k in keys:
    if k == "portfolio":
        print(f"{'metric':<32} {'GMV(raw)':>16} {'GMV(cleaned)':>16}")
        print("-" * 66)
        continue
    # The cross-cov keys differ between m_raw / m_cleaned; align by ordinal
    pass


# Build a parallel printout
def fmt(v):
    if isinstance(v, float):
        return f"{v:.4e}" if abs(v) < 0.01 or abs(v) > 1e3 else f"{v:.4f}"
    return str(v)


labels_and_keys = [
    ("ex-ante variance (own cov)", "variance under raw cov", "variance under␣
↪cleaned cov"),
    ("ex-ante variance (other cov)", "variance under cleaned cov", "variance␣
↪under raw cov"),
    ("gross leverage Σ|w|", "gross leverage Σ|w|", "gross leverage Σ|w|"),
    ("max |w|", "max |w|", "max |w|"),
    ("effective # names", "effective # names 1/Σw²", "effective # names 1/Σw²"),
    ("# negative weights", "# negative weights", "# negative weights"),
]
print(f"{'metric':<32} {'GMV(raw)':>16} {'GMV(cleaned)':>16}")
print("-" * 66)
for label, k_r, k_c in labels_and_keys:
    print(f"{label:<32} {fmt(m_raw[k_r]):>16} {fmt(m_cleaned[k_c]):>16}")

print()
print("Interpretation:")
print(f"  - GMV(raw) in-sample variance is lower than GMV(cleaned) in-sample by␣
↪construction,")
print(f"    but that's the variance of an over-fit portfolio. The cross-cov␣
↪columns are the")
print(f"    informative, 'out-of-sample' values: GMV(cleaned) evaluated under␣
↪Σ_raw is the honest representation of the cleaned portfolio's superiority.")
print(f"  - Gross leverage drops by {m_raw['gross leverage Σ|w|'] /␣
↪m_cleaned['gross leverage Σ|w|']:.1f}x after cleaning")
```

```
print(f"  - the raw inverse cov matrix blows up small eigs, producing extreme␣
↪long-short weights.")
```

```
metric                                   GMV(raw)     GMV(cleaned)
-------------------------------------------------------------------
metric                                   GMV(raw)     GMV(cleaned)
-------------------------------------------------------------------
ex-ante variance (own cov)             3.8243e-07       7.9909e-06
ex-ante variance (other cov)           7.5382e-04       9.1018e-06
gross leverage Σ|w|                       68.2954           6.4780
max |w|                                    0.8459           0.0645
effective # names                          0.0548           7.2481
# negative weights                           237              222

Interpretation:
  - GMV(raw) in-sample variance is lower than GMV(cleaned) in-sample by
construction,
    but that's the variance of an over-fit portfolio. The cross-cov columns are
the
    informative, 'out-of-sample' values: GMV(cleaned) evaluated under Σ_raw is
the honest representation of the cleaned portfolio's superiority.
  - Gross leverage drops by 10.5x after cleaning
  - the raw inverse cov matrix blows up small eigs, producing extreme long-short
weights.
```

11 PC1 - PC5 Eigenportfolios (Ported from John's OOP File)

[50]:
```
def eigenportfolio(evec, normalize="l1"):
    """
    Purpose: Turn eigenvector into portfolio weight vector by normalizing it

    Inputs: eigenvector, normalization method

    Outputs: normalize eigenvector (portfolio weight)
    """
    v = evec.copy()
    if normalize == "l1":
        denom = np.abs(v).sum()
    elif normalize == "l2":
        denom = np.linalg.norm(v)
    elif normalize == "sum1":
        denom = v.sum()
    elif normalize is None:
        denom = 1.0
    else:
        raise ValueError(normalize)
    return v / denom if denom != 0 else v
```

```python
N_PCS = 5
eigenport_weights = np.column_stack([eigenportfolio(evecs[:, i], "l1") for i in␣
↪range(N_PCS)])
eigenport_df = pd.DataFrame(
    eigenport_weights,
    index=returns_win.columns,
    columns=[f"PC{i+1}" for i in range(N_PCS)],
)
eigenport_df["company"] = [col_to_company.get(c, "?") for c in eigenport_df.
↪index]

# Show top-5 long and top-5 short per component
print("Top 5 long and top 5 short positions per eigenportfolio:\n")
for i in range(N_PCS):
    pc = f"PC{i+1}"
    series = eigenport_df[pc]
    top_long = series.nlargest(5)
    top_short = series.nsmallest(5)
    print(f"=== {pc}  (variance share = {evals[i] / evals.sum():.1%}, sign of␣
↪net weight: {'+' if series.sum() > 0 else '-'}) ===")
    print("  LONG:")
    for tic, w in top_long.items():
        print(f"    {tic:<12} {w:+.4f}   {col_to_company.get(tic, '?')}")
    print("  SHORT:")
    for tic, w in top_short.items():
        print(f"    {tic:<12} {w:+.4f}   {col_to_company.get(tic, '?')}")
    print()
```

```
Top 5 long and top 5 short positions per eigenportfolio:

=== PC1  (variance share = 30.1%, sign of net weight: -) ===
  LONG:
    AEP          -0.0000   AMERICAN ELECTRIC POWER CO INC
    WEC          -0.0001   W E C ENERGY GROUP INC
    CMS          -0.0001   C M S ENERGY CORP
    XEL          -0.0001   X C E L ENERGY INC
    DUK          -0.0001   DUKE ENERGY CORP NEW
  SHORT:
    AMD          -0.0047   ADVANCED MICRO DEVICES INC
    NVDA         -0.0045   NVIDIA CORP
    URI          -0.0044   UNITED RENTALS INC
    MU           -0.0044   MICRON TECHNOLOGY INC
    IPGP         -0.0040   I P G PHOTONICS CORP

=== PC2  (variance share = 4.5%, sign of net weight: +) ===
```

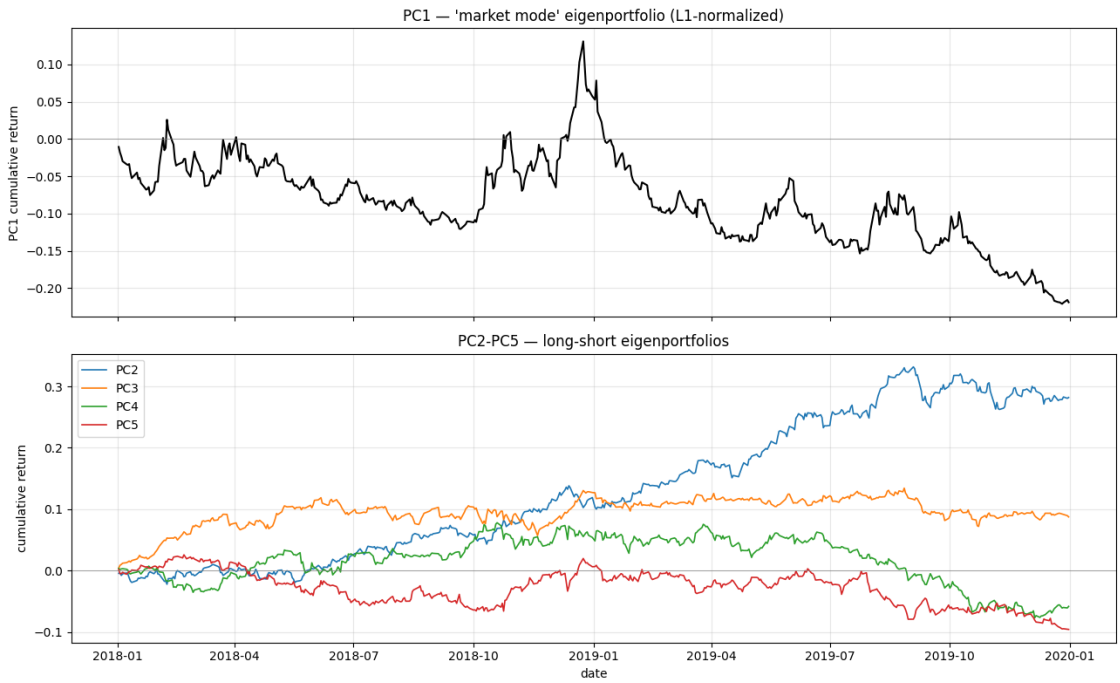
```
  LONG:
    WELL          +0.0056   WELLTOWER INC
    VTR           +0.0055   VENTAS INC
    AMT           +0.0050   AMERICAN TOWER CORP NEW
    O             +0.0049   REALTY INCOME CORP
    SBAC          +0.0049   S B A COMMUNICATIONS CORP NEW
  SHORT:
    APA           -0.0068   A P A CORP
    XEC           -0.0065   CIMAREX ENERGY CO
    DVN           -0.0063   DEVON ENERGY CORP NEW
    MRO           -0.0063   MARATHON OIL CORP
    NBL           -0.0063   NOBLE ENERGY INC

=== PC3  (variance share = 3.0%, sign of net weight: -) ===
  LONG:
    AMD           +0.0093   ADVANCED MICRO DEVICES INC
    NOW_13447     +0.0086   SERVICENOW INC
    NVDA          +0.0083   NVIDIA CORP
    MU            +0.0076   MICRON TECHNOLOGY INC
    ANET_14714    +0.0072   ARISTA NETWORKS INC
  SHORT:
    MS_82621      -0.0133   MILESTONE SCIENTIFIC INC
    HBI_91416     -0.0057   HANESBRANDS INC
    CPB           -0.0050   CAMPBELLS CO
    JWN           -0.0050   NORDSTROM INC
    TAP_59248     -0.0049   MOLSON COORS BEVERAGE CO

=== PC4  (variance share = 2.6%, sign of net weight: +) ===
  LONG:
    MRO           +0.0132   MARATHON OIL CORP
    DVN           +0.0131   DEVON ENERGY CORP NEW
    APA           +0.0125   A P A CORP
    HES           +0.0117   HESS CORP
    XEC           +0.0113   CIMAREX ENERGY CO
  SHORT:
    MS_82621      -0.0547   MILESTONE SCIENTIFIC INC
    AAL           -0.0089   AMERICAN AIRLINES GROUP INC
    WM_81593      -0.0085   WASHINGTON MUTUAL INC
    CPRI          -0.0070   CAPRI HOLDINGS LTD
    UA            -0.0068   UNDER ARMOUR INC

=== PC5  (variance share = 2.3%, sign of net weight: -) ===
  LONG:
    UAA           +0.0089   UNDER ARMOUR INC
    UA            +0.0083   UNDER ARMOUR INC
    AAL           +0.0069   AMERICAN AIRLINES GROUP INC
    HBAN          +0.0051   HUNTINGTON BANCSHARES INC
    ALK           +0.0050   ALASKA AIRGROUP INC
```

```
SHORT:
  MS_82621      -0.0551   MILESTONE SCIENTIFIC INC
  APA           -0.0134   A P A CORP
  HES           -0.0126   HESS CORP
  MRO           -0.0123   MARATHON OIL CORP
  DVN           -0.0123   DEVON ENERGY CORP NEW
```

12 Cumulative Returns from Eigenportfolios

[51]:
```
# Apply L1-normalized weights to returns. eigenport_weights is (p, N_PCS).
daily_returns = X_raw @ eigenport_weights  # (n,p) @ (n, N_PCS) = (n, N_PCS)
cum_returns = np.cumprod(1 + daily_returns, axis=0) - 1
cum_df = pd.DataFrame(cum_returns, index=returns_win.index, columns=[f"PC{i+1}"␣
↪for i in range(N_PCS)])

fig, axes = plt.subplots(2, 1, figsize=(13, 8), sharex=True)

# PC1 alone - has very different scale than the long-short factors
axes[0].plot(cum_df.index, cum_df["PC1"], color="black", linewidth=1.5)
axes[0].axhline(0, color="gray", linewidth=0.5)
axes[0].set_ylabel("PC1 cumulative return")
axes[0].set_title("PC1 - 'market mode' eigenportfolio (L1-normalized)")
axes[0].grid(alpha=0.3)

# PC2-PC5 together
for col in ["PC2", "PC3", "PC4", "PC5"]:
    axes[1].plot(cum_df.index, cum_df[col], label=col, linewidth=1.2)
axes[1].axhline(0, color="gray", linewidth=0.5)
axes[1].set_ylabel("cumulative return")
axes[1].set_xlabel("date")
axes[1].set_title("PC2-PC5 - long-short eigenportfolios")
axes[1].legend()
axes[1].grid(alpha=0.3)

plt.tight_layout()
plt.show()

# Summary stats
ann_factor = 252
print("\nAnnualized stats per eigenportfolio (window = 2018-2019):\n")
stats = pd.DataFrame(
    {
        "ann_return": daily_returns.mean(axis=0) * ann_factor,
        "ann_vol": daily_returns.std(axis=0) * np.sqrt(ann_factor),
        "sharpe": (daily_returns.mean(axis=0) / daily_returns.std(axis=0)) * np.
↪sqrt(ann_factor),
```

```
        "var_share": [evals[i] / evals.sum() for i in range(N_PCS)],
    },
    index=[f"PC{i+1}" for i in range(N_PCS)],
)
print(stats.to_string(float_format=lambda x: f"{x:.4f}"))
```




```
Annualized stats per eigenportfolio (window = 2018-2019):

     ann_return  ann_vol  sharpe  var_share
PC1     -0.1098   0.1672 -0.6565     0.3014
PC2      0.1271   0.0726  1.7498     0.0446
PC3      0.0439   0.0616  0.7126     0.0299
PC4     -0.0267   0.0822 -0.3251     0.0255
PC5     -0.0474   0.0777 -0.6108     0.0226
```

13 Rolling Assessment of GMV Portfolios

[52]:
```
WIN_DAYS = 504  # ~2 trading years - keeps q = p/n < 1 so MP and matrix␣
↪inversion are well-posed (we learned this the hard way when tracking␣
↪dimensionality throughout time)
STEP_DAYS = 63
records_v2 = []
dates = returns_full.index
```

```
for start_idx in range(0, len(dates) - WIN_DAYS + 1, STEP_DAYS):
    end_idx = start_idx + WIN_DAYS
    end_date = dates[end_idx - 1]
    members_w = members_as_of(end_date)
    cols_w = members_to_columns(members_w)
    win = returns_full.iloc[start_idx:end_idx][cols_w].dropna(axis=1, how="any")
    n_w, p_w = win.shape
    if p_w < 20 or p_w >= n_w:
        continue

    Xw = win.to_numpy(dtype=float)
    Xw = Xw - Xw.mean(axis=0, keepdims=True)
    cov_w = (Xw.T @ Xw) / (n_w - 1)
    ev_w, vec_w = eigh(cov_w)
    ev_w = ev_w[::-1]
    vec_w = vec_w[:, ::-1]
    q_w = p_w / n_w

    s2_w = iter_sigma2(ev_w, q_w)
    edge_w = s2_w * (1 + np.sqrt(q_w)) ** 2
    cov_clean_w, ev_clean_w = rmt_clean_covariance(ev_w, vec_w, edge_w)

    try:
        w_r = gmv_weights(cov_w)
        w_c = gmv_weights(cov_clean_w)
        gross_r = float(np.abs(w_r).sum())
        gross_c = float(np.abs(w_c).sum())
    except np.linalg.LinAlgError:
        gross_r = gross_c = np.nan

    pos_raw = ev_w[ev_w > 0]
    cond_raw_w = float(pos_raw.max() / pos_raw.min()) if len(pos_raw) > 0 else␣
↪np.nan
    cond_cleaned_w = float(ev_clean_w.max() / ev_clean_w.min())

    records_v2.append(
        {
            "window_end": end_date,
            "n_assets": p_w,
            "q": q_w,
            "rmt_dim": int((ev_w > edge_w).sum()),
            "top_eig_share": ev_w[0] / ev_w.sum(),
            "cond_raw": cond_raw_w,
            "cond_cleaned": cond_cleaned_w,
            "gross_raw": gross_r,
            "gross_cleaned": gross_c,
```

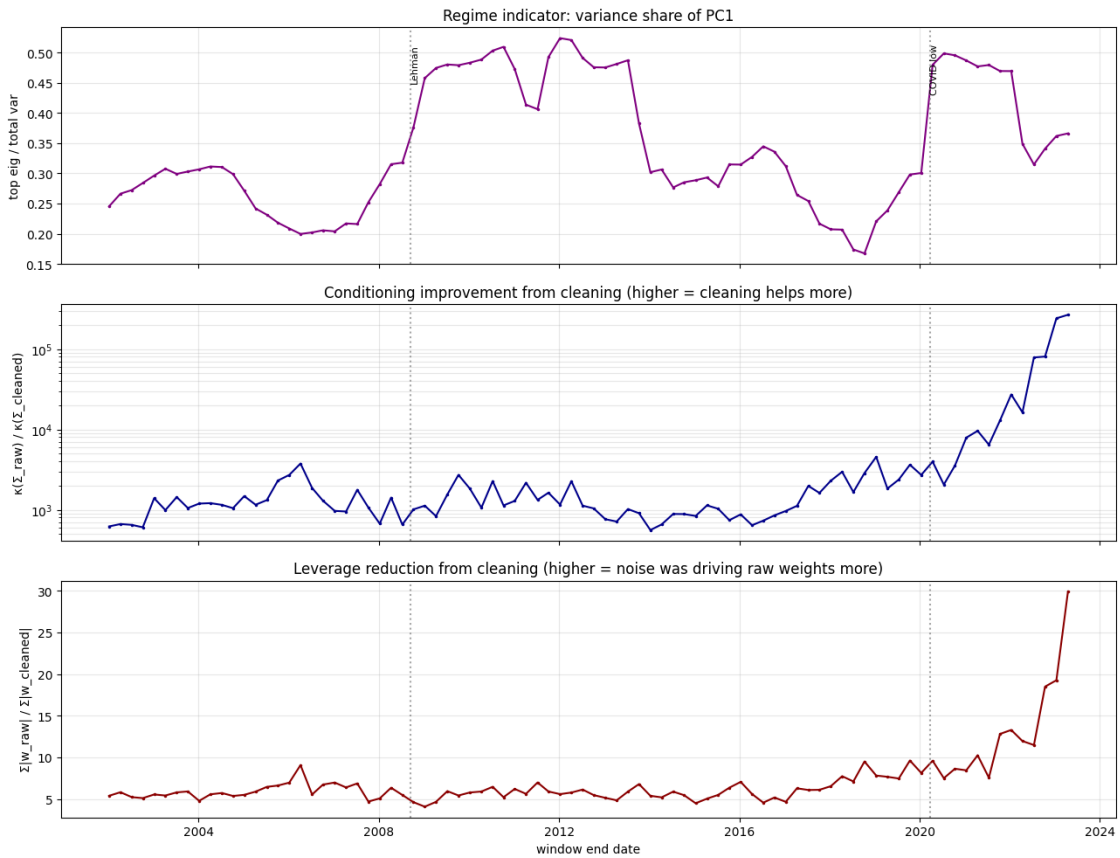
```
        }
    )

rolling_v2 = pd.DataFrame(records_v2).set_index("window_end")
rolling_v2["cond_ratio"] = rolling_v2["cond_raw"] / rolling_v2["cond_cleaned"]
rolling_v2["gross_ratio"] = rolling_v2["gross_raw"] /␣
↪rolling_v2["gross_cleaned"]
print(f"Computed {len(rolling_v2)} rolling windows ({WIN_DAYS}-day window).")
print(f"q range across windows: [{rolling_v2['q'].min():.3f}, {rolling_v2['q'].
↪max():.3f}]")
rolling_v2[["n_assets", "q", "rmt_dim", "top_eig_share", "cond_ratio",␣
↪"gross_ratio"]].head()
```

```
Computed 86 rolling windows (504-day window).
q range across windows: [0.935, 0.996]
```

[52]:

window_end	n_assets	q	rmt_dim	top_eig_share	cond_ratio	gross_ratio
2002-01-07	472	0.9365	139	0.2453	620.6438	5.3872
2002-04-09	474	0.9405	155	0.2663	664.0463	5.8157
2002-07-09	478	0.9484	154	0.2722	649.0970	5.2271
2002-10-07	478	0.9484	162	0.2840	605.1685	5.1074
2003-01-07	481	0.9544	162	0.2963	1393.5763	5.5442

[53]:

```
fig, axes = plt.subplots(3, 1, figsize=(13, 10), sharex=True)

axes[0].plot(rolling_v2.index, rolling_v2["top_eig_share"], color="purple",␣
↪marker=".", markersize=3)
axes[0].set_ylabel("top eig / total var")
axes[0].set_title("Regime indicator: variance share of PC1")
axes[0].grid(alpha=0.3)

axes[1].plot(rolling_v2.index, rolling_v2["cond_ratio"], color="darkblue",␣
↪marker=".", markersize=3)
axes[1].set_yscale("log")
axes[1].set_ylabel(" (Σ_raw) /  (Σ_cleaned)")
axes[1].set_title("Conditioning improvement from cleaning (higher = cleaning␣
↪helps more)")
axes[1].grid(alpha=0.3, which="both")

axes[2].plot(rolling_v2.index, rolling_v2["gross_ratio"], color="darkred",␣
↪marker=".", markersize=3)
axes[2].set_ylabel("Σ|w_raw| / Σ|w_cleaned|")
axes[2].set_xlabel("window end date")
axes[2].set_title("Leverage reduction from cleaning (higher = noise was driving␣
↪raw weights more)")
axes[2].grid(alpha=0.3)
```

```
for date_str, label in [("2008-09-15", "Lehman"), ("2020-03-23", "COVID low")]:
    d = pd.Timestamp(date_str)
    if rolling_v2.index.min() <= d <= rolling_v2.index.max():
        for ax in axes:
            ax.axvline(d, color="gray", linestyle=":", alpha=0.7)
        axes[0].text(d, axes[0].get_ylim()[1] * 0.95, label, rotation=90,␣
 ↪va="top", fontsize=8)

plt.tight_layout()
plt.show()
```